Lab Manual for Digital Logic and Design

LAB-11

Implementation of BCD to seven segment conversion

EXPERIMENT NO. : 08

Implementation of BCD to seven segment conversion

Objective:

To familiarize the student with the seven-segment LED display, and the process of converting one type of binary information to another (encoding/decoding).

Apparatus:

Multisim software/ trainer
+5 V, 0 V input voltage
Resistors 220 Ω
IC7447
Seven segment LED display

Description:

The seven-segment LED (Light Emitting Diode) display is a common device in consumer electronics, from calculators to clocks to microwave ovens. This lab includes basic principles of operation of the seven-segment display and the process of converting BCD values to the proper signals to drive this display. The display has seven separate bar-shaped LED's arranged as shown.

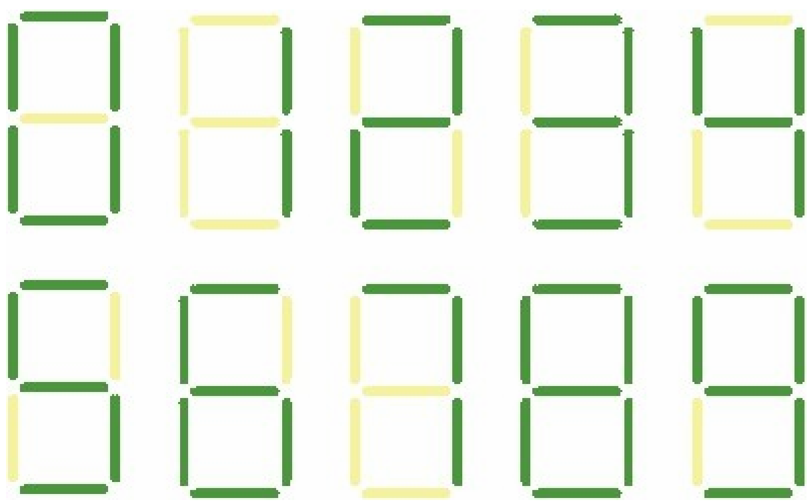


Figure 8.1. Seven Segment Display Format

Inside the seven-segment display, one end of each LED is connected to a common point. This common point is tied either to ground or to the positive supply, depending on the specific device. If your seven segment display is designed to have the common connection tied to the positive supply, +5V, it is called a common anode configuration as shown below. To light these LED segments, the inputs must be a logic low

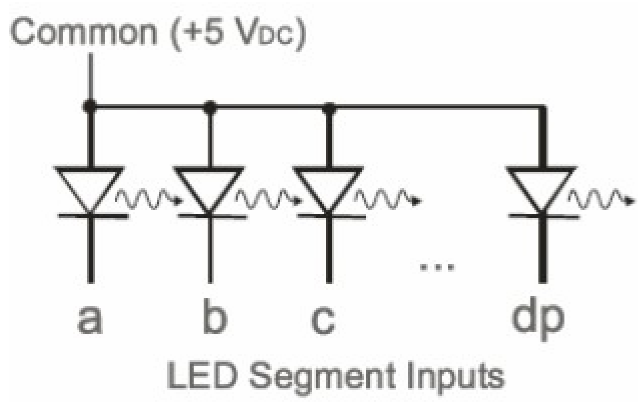


Figure 8.2. Common Anode Seven-Segment Display Circuitry

To actually light up a single LED segment, a resistor must be added to limit the current through the LED. This resistor is critical. If you connect the LED between +5V and ground without the resistor, the LED will momentarily glow bright and then never glow again. For this display use, a 220-Ω resistor as shown below

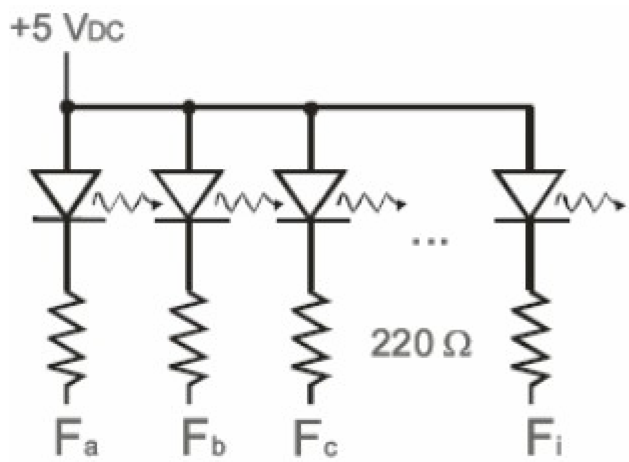


Figure 8.3. Resistors used with a Common Anode Seven-Segment Display

If Fa, Fb, etc... are +5 V, there is no voltage drop across the LED and resister resulting in no current flow through them, (and the LED remains dark.) If the inputs are 0 Volts, a current is produced and the diode glows.

In a common cathode display, the common point is ground instead of +5 V as shown below. To light these segments, a logic high must be supplied.

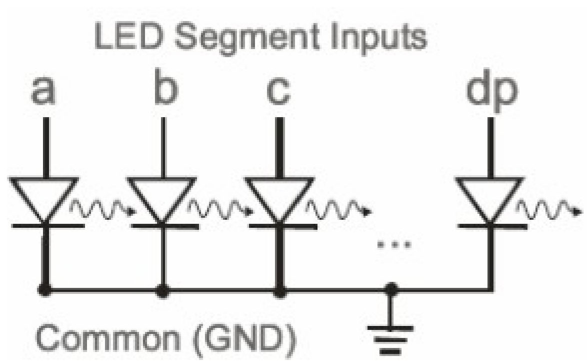


Figure 8.4. Common Cathode Seven-Segment Display Circuitry

IC7447 is called a BCD-to-seven-segment display which is shown below

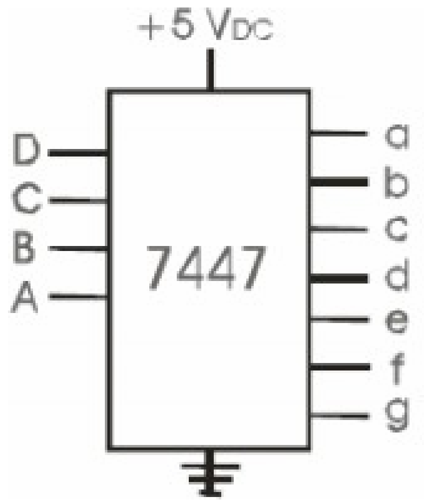


Figure 8.5. 7447 BCD to Seven-Segment Display

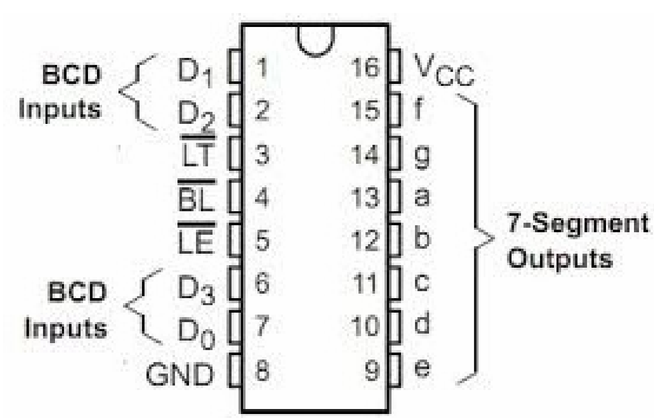


Figure 8.6. IC7447 pin configuration

Table 8.1: Pin configuration with discriotion

PIN NUMBER	DESCRIPTION
1	BCD B Input
2	BCD C Input
3	Lamp Test
4	RB Output
5	RB Input
6	BCD D Input
7	BCD A Input
8	Ground
9	7-Segment e Output
10	7-Segment d Output
11	7-Segment c Output
12	7-Segment b Output
13	7-Segment a Output
14	7-Segment g Output
15	7-Segment f Output
16	Positive Supply

BCD to 7 Segment Display Decoder

A BCD to Seven Segment decoder is a combinational logic circuit that accepts a decimal digit in BCD (input) and generates appropriate outputs for the segments to display the input decimal digit.

Circuit Connection:

Connect the circuit as shown below using IC7447 and 7 Segment Display and verify the truth table given below.

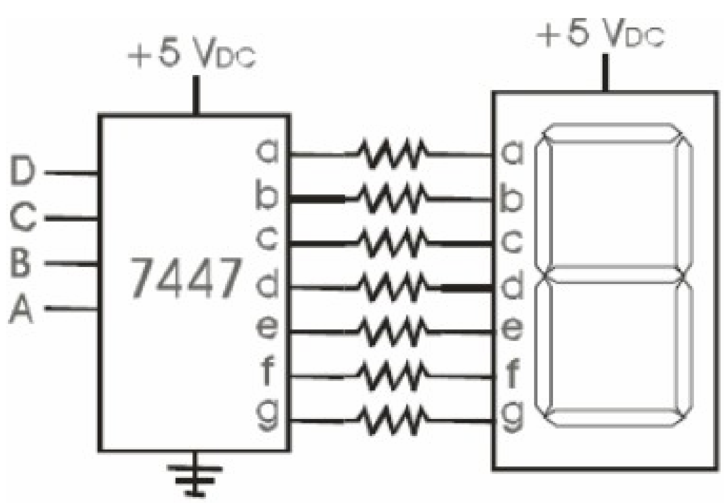
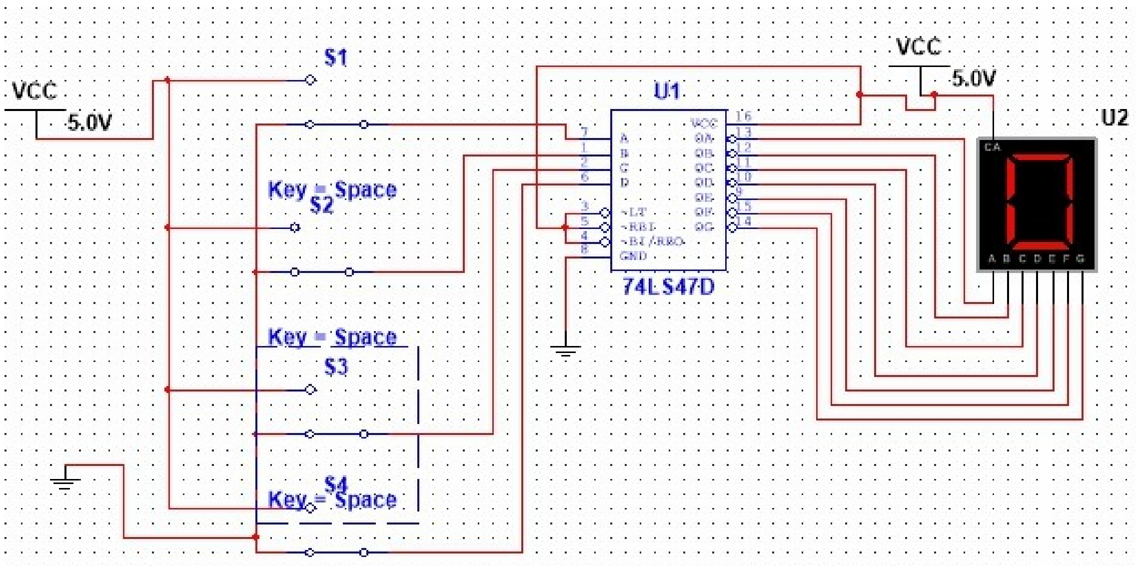
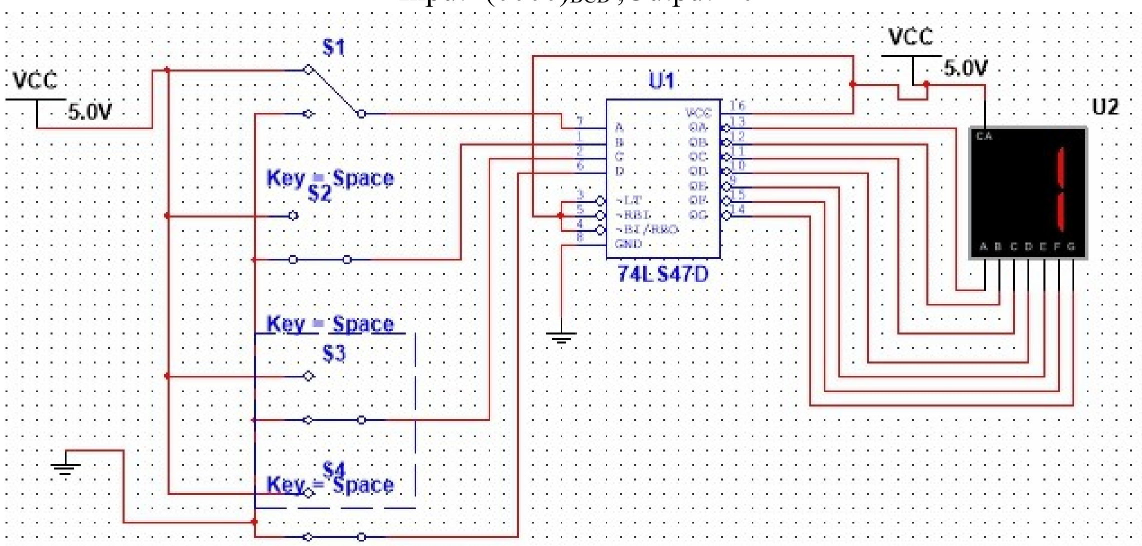


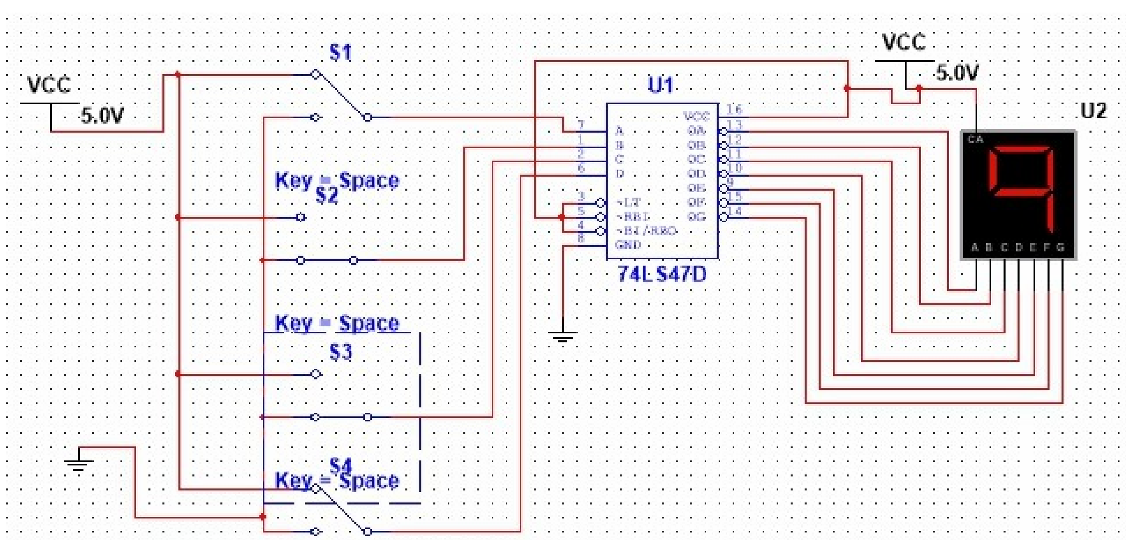
Figure 8.7: BCD to 7 segment display



Input=$(0000)_{BCD}$,Output= 0



Input=$(0001)_{BCD}$,Output= 1



Input=$(1001)_{BCD}$,Output= 9

Table 8.2: Truth Table of Seven Segment Decoder

Binary Inputs				**Decoder Outputs**							**7-Segment Display Outputs**
A	**B**	**C**	**D**	**a**	**b**	**c**	**d**	**e**	**f**	**g**	
0	**0**	**0**	**0**	**1**	**1**	**1**	**1**	**1**	**1**	**0**	**0**
0	**0**	**0**	**1**	**0**	**1**	**1**	**0**	**0**	**0**	**0**	**1**
0	**0**	**1**	**0**	**1**	**1**	**0**	**1**	**1**	**0**	**1**	**2**
0	**0**	**1**	**1**	**1**	**1**	**1**	**1**	**0**	**0**	**1**	**3**
0	**1**	**0**	**0**	**0**	**1**	**1**	**0**	**0**	**1**	**1**	**4**
0	**1**	**0**	**1**	**1**	**0**	**1**	**1**	**0**	**1**	**1**	**5**
0	**1**	**1**	**0**	**1**	**0**	**1**	**1**	**1**	**1**	**1**	**6**
0	**1**	**1**	**1**	**1**	**1**	**1**	**0**	**0**	**0**	**0**	**7**
1	**0**	**0**	**0**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**8**
1	**0**	**0**	**1**	**1**	**1**	**1**	**1**	**0**	**1**	**1**	**9**

Conclusion:

Implemented the following circuit in MultiSim. The outputs were observed on LED's andoutputs were same as truth table.